

The Linear Regression Model

Part I — Introduction & OLS Estimation

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Lecture Overview

Lecture 8 — Introduction & OLS Estimation

- Why regression? Motivation and examples
- Statistical vs. deterministic relationships
- Scatter diagrams and linear correlation
- The Simple Linear Regression (SLR) model
- The Classical Assumptions (Gauss–Markov)
- OLS: deriving $\hat{\beta}_0$ and $\hat{\beta}_1$
- Interpreting the estimated coefficients
- Running a regression in R with `lm()`

Why Regression?

Motivation

In many situations, we observe that variables tend to **move together**:

Accounting & Finance

- Sales revenue → Operating profit
- Advertising spend → Market share
- Firm leverage → Cost of debt
- Total assets → Audit fees

Economics & Management

- Disposable income → Consumption
- Interest rates → Investment
- Education → Earnings
- Hours studied → Exam grade

Regression analysis lets us:

1. **Describe** the average relationship between variables

2. **Quantify** the size of that relationship

Statistical vs. Deterministic Relationships

Deterministic:

$$C = 2\pi r$$

Every value of r gives **exactly one** C .

No randomness. No need for statistics.

Statistical:

$$\text{Profit}_i = \beta_0 + \beta_1 \text{Revenue}_i + \varepsilon_i$$

Same revenue $\rightarrow$ **different** profit across firms.

The error ε_i captures everything else.

! Important

Regression models the **conditional mean** of Y given X :

$$E[Y \mid X = x] = \beta_0 + \beta_1 x$$

We do **not** claim X determines Y **exactly** — **only on average**

Introduction & OLS

Types of Variables

Role	Name	Also called	Symbol
What we explain	Dependent variable	Response, regressand	Y
What explains it	Independent variable	Regressor, predictor	X

- Y is always **quantitative**.
- X can be quantitative or qualitative (dummy variables — introduced in Part II).

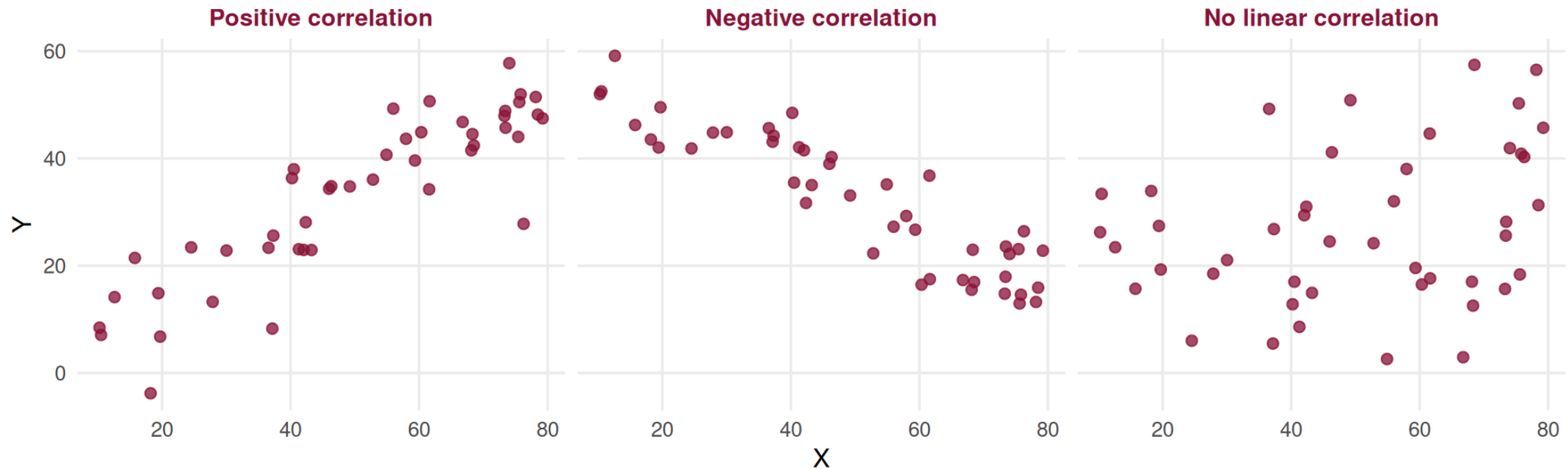
Running example (hypothetical data throughout):

A firm's operating profit (Y , m.u.) as a function of sales revenue (X , m.u.).

Exploring the Relationship

The Scatter Diagram

Before fitting any model, always **plot the data**. The scatter diagram displays the n observed pairs (x_i, y_i) .



Reading the Scatter Diagram

From the scatter diagram we assess:

- **Direction:** positive (both increase together) or negative (one increases, the other decreases)?
- **Form:** linear, curved, or no discernible pattern?
- **Strength:** how tightly do points cluster around the trend?
- **Outliers:** any unusual observations that break the pattern?

! Important

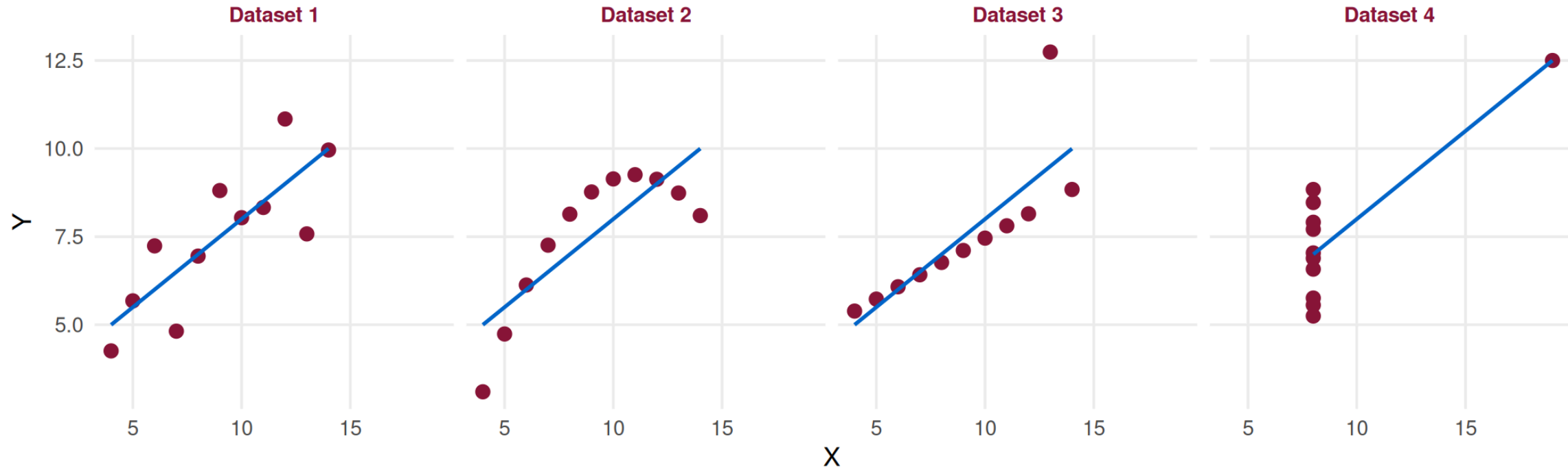
Always plot before you fit.

A high R^2 on a non-linear or structurally misspecified relationship can be meaningless. Classic illustration:

Anscombe's Quartet.

Anscombe's Quartet

All four datasets: same slope, intercept, and R^2



$\hat{\beta}_0 \approx 3.0$, $\hat{\beta}_1 \approx 0.50$, $R^2 \approx 0.67$ for **all four** — but only Dataset 1 is truly linear.

The SLR Model

Model Specification

The **Simple Linear Regression (SLR)** model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, 2, \dots, n$$

Symbol	Name	Meaning
Y_i	Dependent variable	Observed outcome for unit i
X_i	Independent variable	Known regressor for unit i
β_0	Intercept	Mean of Y when $X = 0$
β_1	Slope	Change in mean Y per unit increase in X
ε_i	Error term	All other factors affecting Y_i

The Error Term ε_i

$$Y_i = \underbrace{\beta_0 + \beta_1 X_i}_{\text{systematic part}} + \underbrace{\varepsilon_i}_{\text{random part}}$$

The error captures:

- **Omitted variables** — factors that influence Y but are not in the model
- **Measurement error** — imprecision in recording Y_i or X_i
- **Inherent randomness** — unpredictable variation in behaviour or outcomes

⚠ Important

The error is **not a mistake** in the model — it is an unavoidable feature of any statistical relationship. What matters is that we make appropriate assumptions about its behaviour & OLS

The Classical Assumptions

Gauss–Markov Assumptions

For OLS to have good statistical properties, we require:

#	Name	Statement
A1	Linearity	$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$
A2	Zero mean	$E[\varepsilon_i] = 0 \forall i$
A3	Homoscedasticity	$\text{Var}(\varepsilon_i) = \sigma^2 \forall i$
A4	No autocorrelation	$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \forall i \neq j$
A5	Exogeneity	X_i is fixed or independent of ε_i
A6	Normality	$\varepsilon_i \sim N(0, \sigma^2)$

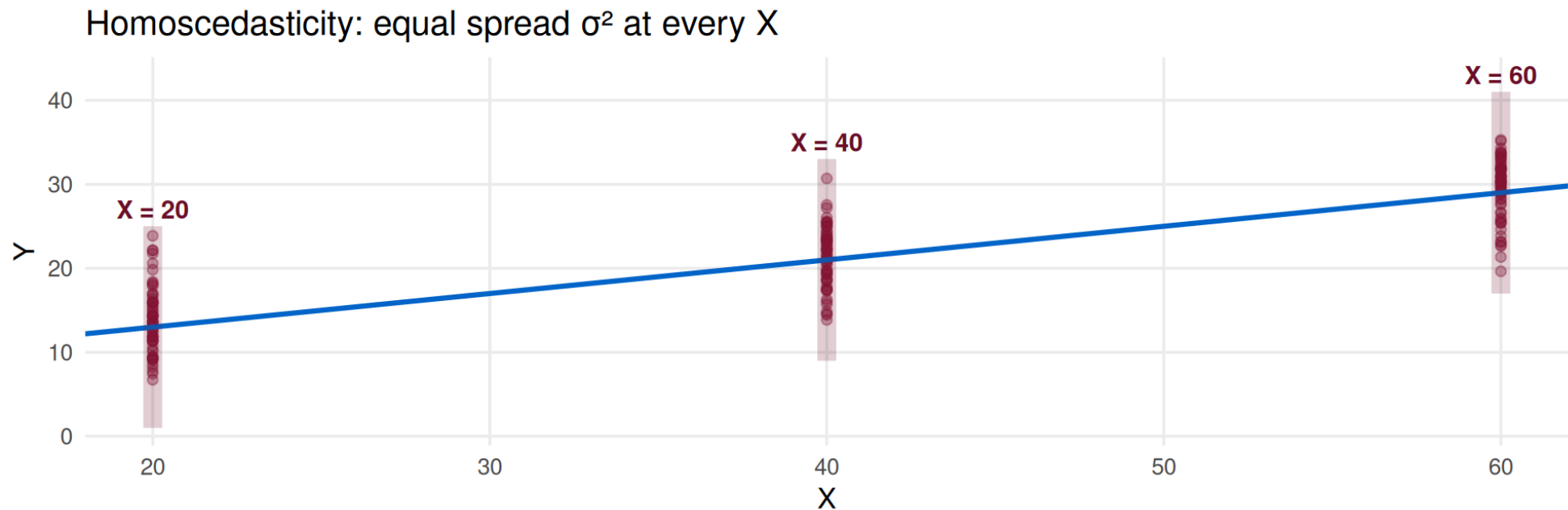
Note

A1–A5 are the Gauss–Markov conditions. Under them, OLS is **BLUE** (Best Linear Unbiased Estimator). **A6** is additionally required for exact t - and F -tests in small samples.

What the Assumptions Imply

Under A1–A6, for each i :

$$Y_i \mid X_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$$



The conditional distribution of $Y|X$ is Normal with the same variance at every X value

OLS Estimation

The Idea: Minimise Squared Residuals

Given data $(x_1, y_1), \dots, (x_n, y_n)$, the **fitted line** is:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

The **residual** for observation i :

$$e_i = Y_i - \hat{Y}_i$$

! Important

Ordinary Least Squares (OLS): choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimise the **Sum of Squared Residuals**:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \text{SSR} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

OLS: Visualising the Criterion



OLS Formulas

Taking partial derivatives of SSR and setting them to zero yields:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{S_{XY}}{S_{XX}}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

where $S_{XY} = \sum (X_i - \bar{X})(Y_i - \bar{Y})$ and $S_{XX} = \sum (X_i - \bar{X})^2$.

Deriving the OLS Formulas

Minimise $SSR = \sum (Y_i - \beta_0 - \beta_1 X_i)^2$:

$$\frac{\partial SSR}{\partial \beta_0} = -2 \sum (Y_i - \beta_0 - \beta_1 X_i) = 0$$

$$\frac{\partial SSR}{\partial \beta_1} = -2 \sum X_i (Y_i - \beta_0 - \beta_1 X_i) = 0$$

These are the **Normal Equations**:

$$\sum Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum X_i$$

$$\sum X_i Y_i = \hat{\beta}_0 \sum X_i + \hat{\beta}_1 \sum X_i^2$$

Interpreting the Estimates

Interpreting $\hat{\beta}_1$ and $\hat{\beta}_0$

The **estimated regression equation**:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

Slope $\hat{\beta}_1$:

For every **one-unit increase** in X , the estimated mean of Y changes by $\hat{\beta}_1$ units.

This is a **marginal effect** — the effect of X on the average Y .

Intercept $\hat{\beta}_0$:

The estimated mean of Y when $X = 0$.

Often not directly meaningful — $X = 0$ may be outside the range of the data.

R Demo: Fitting and Interpreting

Hypothetical data: 20 observations on sales revenue (X) and operating profit (Y), both in m.u.

```
1 set.seed(5)
2 n      <- 20
3 revenue <- round(runif(n, 100, 500), 0)
4 profit  <- round(10 + 0.18 * revenue + rnorm(n, 0, 15), 1)
5
6 model <- lm(profit ~ revenue)
7 coef(model)
```

(Intercept)	revenue
21.8268848	0.1435997

Interpretation:

- $\hat{\beta}_1 = 0.144$: each additional m.u. of revenue is associated with an average increase of 0.144 m.u. in operating profit.

R Demo: The Full Summary

```
1 summary(model)
```

Call:

```
lm(formula = profit ~ revenue)
```

Residuals:

Min	1Q	Median	3Q	Max
-29.169	-10.033	-3.226	11.177	27.235

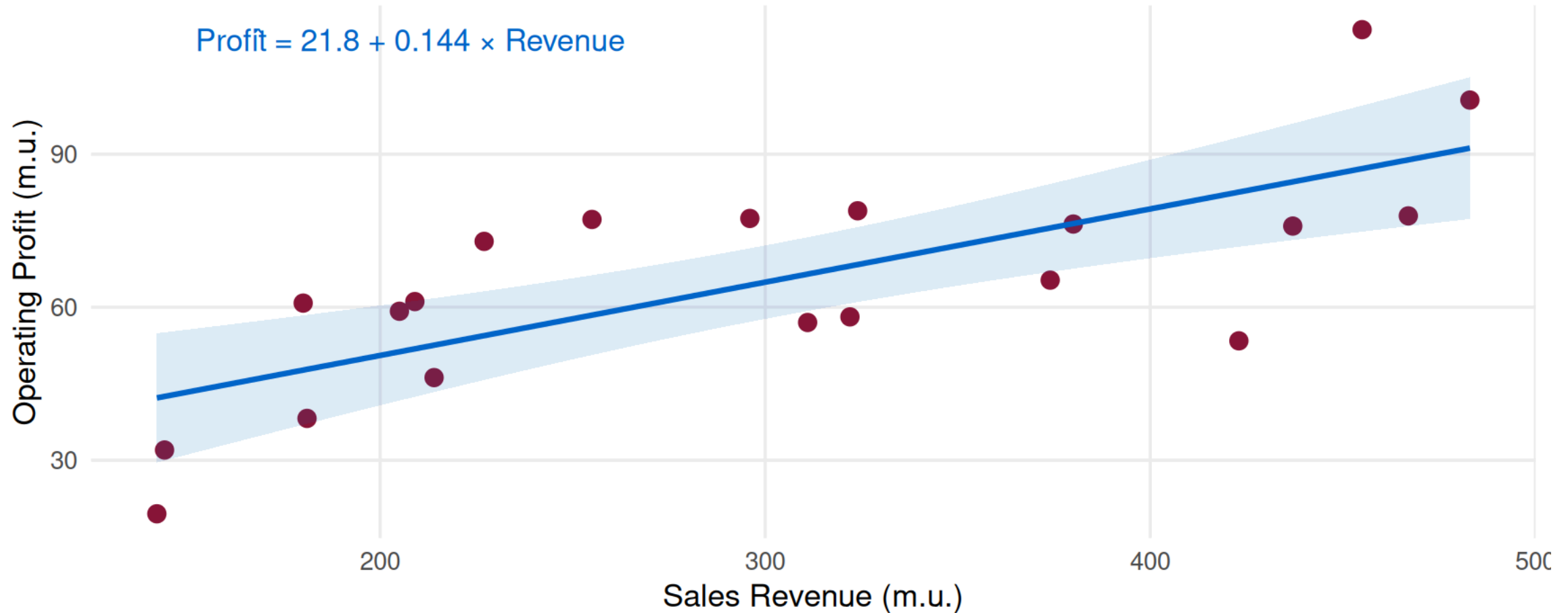
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	21.82688	9.99146	2.185	0.042392	*
revenue	0.14360	0.03113	4.612	0.000216	***

Signif. codes: 0. '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R Demo: The Fitted Line

Operating Profit vs. Sales Revenue (hypothetical data)



Properties of OLS Residuals

These **algebraic identities** hold for every OLS fit:

$$\sum_{i=1}^n e_i = 0$$

$$\sum_{i=1}^n X_i e_i = 0$$

$$\sum_{i=1}^n \hat{Y}_i e_i = 0$$

They are not assumptions — they are guaranteed by the OLS first-order conditions. The residuals automatically sum to zero and are uncorrelated with X and with $\hat{Y}$.

Estimating the Error Variance

The population error variance σ^2 is unknown. We estimate it with:

$$\hat{\sigma}^2 = s^2 = \frac{\text{SSR}}{n - 2} = \frac{\sum_{i=1}^n e_i^2}{n - 2}$$

We divide by $n - 2$ because estimating $\hat{\beta}_0$ and $\hat{\beta}_1$ **uses up two degrees of freedom**.

The **standard error of the regression** (residual standard error in R):

$$s = \sqrt{\frac{\sum e_i^2}{n - 2}}$$

Lecture Summary

Key Takeaways — Lecture 8

- **Regression** models the **conditional mean** of Y given X — a statement about averages, not individual values.
- The **SLR model** is $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$; the error ε_i is unavoidable.
- The **Gauss–Markov assumptions** (A1–A5) ensure OLS is BLUE; adding normality (A6) enables exact inference.
- **OLS** chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimise $\sum e_i^2$:

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

- $\hat{\beta}_1$ is the **marginal effect**: estimated change in mean Y per unit increase in X .

Looking Ahead

Lecture 9 — Part II: Goodness of Fit, OLS Properties & Introduction to MLR

- How well does the line fit? — R^2 and the variance decomposition
- Why trust OLS? — The Gauss–Markov theorem
- What if we leave out a relevant variable? — **Omitted Variable Bias**
- Adding more regressors — **Multiple Linear Regression**

 Note

Reference: Newbold Ch. 11.4–11.5, Ch. 12.1–12.3.